

Notes on differentiating through the HPS-based Helmholtz Solver

Olivia Tsang

Owen Melia

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1 Introduction

This note describes the implementation of the vector-Jacobian products implemented in this repository, based on the GPU-accelerated `jaxhps` numerical solver for Hierarchical Poincaré-Steklov (HPS) methods (Melia et al., 2026). This method was developed in Gillman et al. (2015); Borges et al. (2017); see also a reference implementation at <https://github.com/flatironinstitute/inverse-volume-scattering2d>. The point of this note is to provide a detailed description of this method.

Let $q(x)$ denote the scattering potential, and let the forward model at frequency k for source s be denoted as $\mathcal{F}_{k,s}[q]$. We want to compute vector-Jacobian products $\xi_s^* D\mathcal{F}_{k,s}[q] = D\mathcal{F}_{k,s}[q]^* \xi_s$, where ξ_s is a vector supported at the receiver locations and $D\mathcal{F}_{k,s}[q]$ is the Fréchet derivative of $\mathcal{F}_{k,s}$ at q . We will refer to this object as the Jacobian.

1.1 Adjoint PDE

Borges et al. (2017) derives an adjoint equation to compute $D\mathcal{F}_{k,s}[q]^* \xi_s$:

$$(D\mathcal{F}_{k,s}[q]^* \xi_s)(x) = \text{Re} \left[\overline{u_s^{(tot)}(x)} w_s(x) \right], \quad (1)$$

where $\overline{(\cdot)}$ represents complex conjugation and $u_s^{(tot)}(x) = u_s^{(in)}(x) + u_s^{scat}(x)$ is solution of the Helmholtz equation with scattering potential $q(x)$:

$$\begin{cases} \Delta u_s^{(tot)}(x) + k^2(q(x) + 1)u_s^{(tot)}(x) = 0, & x \in \mathbb{R}^2; \\ \frac{\partial}{\partial r} u_s^{(tot)}(x) - iku_s^{(tot)}(x) = o(r^{-1/2}), & \text{as } r = \|x\|_2 \rightarrow \infty; \end{cases} \quad (2)$$

and $w_s(x)$ is the solution to the following adjoint PDE:

$$\begin{cases} \Delta w_s(x) + k^2(q(x) + 1)w_s(x) = -k^2 \chi(\xi_s, \partial B)(x), & x \in \mathbb{R}^2; \\ \frac{\partial}{\partial r} w_s(x) + ikw_s(x) = o(r^{-1/2}), & \text{as } r = \|x\|_2 \rightarrow \infty. \end{cases} \quad (3)$$

Following the notation from Borges et al. (2017), $\chi(\xi_s, \partial B)$ embeds the function $\xi_s : \partial B \rightarrow \mathbb{C}$ (i.e., takes inputs on the receiver ring, with radius R) into $\mathbb{R}^2 \rightarrow \mathbb{C}$ (i.e., takes any inputs in $\mathbb{R}^2$). It is helpful to think of $\chi(\xi_s, \partial B)$ as a sum of Dirac delta functions, centered at the receiver points, with weights ξ_s .

Equation (3) looks very similar to the usual Helmholtz equation, except that there is a nonzero source term supported at a discrete set of receiver locations, and $w_s(x)$ now obeys the *adjoint* Sommerfeld radiation condition (the sign for $ikw_s(x)$ is flipped).

Note that when using multiple wave sources, we can apply Equation (1) to each ξ_s independently, then sum over the sources s and take the real component (since $q(x)$ is real-valued):

$$(D\mathcal{F}_k[q]^* \xi)(x) = \sum_s \text{Re} \left[\overline{u_s^{(tot)}(x)} w_s(x) \right]. \quad (4)$$

For example, we might take $\xi = d_k - \mathcal{F}_k[\hat{q}]$ where $d_k = \mathcal{F}_k[q]$ is the true set of scattering measurements, and $\hat{q}$ is a guess of the scattering potential.

1.2 Decomposing the adjoint PDE

Directly discretizing the source term in Equation (3) is inadvisable due to its support at a discrete set of receiver points. Instead, one can decompose Equation (3) into a constant-coefficient problem with sources at the receiver points and an variable-coefficient problem with a corrected source. We write $w_s(x) = \alpha_s(x) + \beta_s(x)$, where $\alpha_s(x)$ solves the constant-coefficient problem so we can focus on $\beta_s(x)$ on the usual domain of Ω . Given this decomposition, we define α_s such that

$$\begin{cases} (\Delta + k^2)\alpha_s(x) = -k^2\chi(\xi_s, \partial B)(x), & x \in \mathbb{R}^2; \\ \frac{\partial}{\partial r}\alpha_s(x) + ik\alpha_s(x) = o(r^{-1/2}), & r = \|x\|_2 \rightarrow \infty. \end{cases} \quad (5)$$

Solving for α_s is relatively simple. $H_0^{(2)}$, the Hankel function of the second kind and 0^{th} order, is the Green's function for the constant-coefficient Helmholtz equation with adjoint radiation conditions. Thus, $\alpha_s(x)$ is a linear combination of $H_0^{(2)}$ appropriately weighted and centered at the receiver locations. Because the receivers are separated from the computational domain Ω , α_s is smooth on Ω . Using the decomposition $w_s(x) = \alpha_s(x) + \beta_s(x)$, we can derive the following problem for β_s :

$$\begin{cases} \Delta\beta_s(x) + k^2(q(x) + 1)\beta_s(x) = -k^2q(x)\alpha_s(x), & x \in \mathbb{R}^2; \\ \frac{\partial}{\partial r}\beta_s(x) + ik\beta_s(x) = o(r^{-1/2}), & r = \|x\|_2 \rightarrow \infty. \end{cases} \quad (6)$$

2 Solving the adjoint PDE with Dirichlet-to-Impedance operators

2.1 Splitting into particular and homogeneous solutions

We split Equation (6) into particular and homogeneous problems (corresponding to $\beta_s^{(p)}(x)$ and $\beta_s^{(h)}(x)$, respectively), where the particular solution $\beta_s^{(p)}(x)$ solves

$$\begin{cases} \Delta\beta_s^{(p)}(x) + k^2(q(x) + 1)\beta_s^{(p)}(x) = -k^2q(x)\alpha_s(x), & x \in \Omega; \\ \partial_n\beta_s^{(p)}(x) + ik\beta_s^{(p)}(x) = 0, & x \in \partial\Omega; \end{cases} \quad (7)$$

and the homogeneous solution $\beta_s^{(h)}(x)$ solves

$$\begin{cases} \Delta\beta_s^{(h)}(x) + k^2(q(x) + 1)\beta_s^{(h)}(x) = 0, & x \in \Omega; \\ \partial_n\beta_s^{(h)}(x) + ik\beta_s^{(h)}(x) = g_s(x), & x \in \partial\Omega; \end{cases} \quad (8)$$

where $g_s(x)$ is the incoming impedance boundary data. The HPS solver requires boundary data specified as in Equation (8); one must compute appropriate data $g_s(x)$ so that $\beta(x)$ solves Equation (6).

To compute this boundary data, one can leverage the structure of the source term in Equation (7) and, treating α_s as the incident wave and β_s as the scattered wave, use the approach described in (Gillman et al., 2015, Equation 3.4). In the following section, we describe a variant of the method which applies to general source terms.

2.2 Computing boundary data with Dirichlet-to-Impedance operators

We can consider the continuity conditions at the domain boundary, $\partial\Omega$. Let $\partial_n(\cdot)|_{\partial\Omega}$ be the outward-pointing normal derivative of a field at the domain boundary. If $\beta_s|_{\partial\Omega}$ is the solution trace of Equation (6) at $\partial\Omega$ from the interior and $\gamma_s|_{\partial\Omega}$ is the solution trace from the exterior, then we can set up the following equations from the Dirichlet and Impedance boundary conditions:

$$\gamma_s|_{\partial\Omega} = \beta_s|_{\partial\Omega} \quad (9)$$

$$= \beta_s^{(p)}|_{\partial\Omega} + \beta_s^{(h)}|_{\partial\Omega} \quad (10)$$

$$\partial_n\gamma_s|_{\partial\Omega} + ik\gamma_s|_{\partial\Omega} = (\partial_n\beta_s|_{\partial\Omega} + ik\beta_s|_{\partial\Omega}) \quad (11)$$

$$= (\partial_n\beta_s^{(h)}|_{\partial\Omega} + ik\beta_s^{(h)}|_{\partial\Omega}) \quad \text{since } \partial_n\beta_s^{(p)}|_{\partial\Omega} + ik\beta_s^{(p)}|_{\partial\Omega} = 0. \quad (12)$$

Let $G_{int}^{(DtI)}$ be the Dirichlet-to-Impedance map for the solution on the interior of Ω , and let $G_{ext}^{(DtI)}$ be the same for the solution on the exterior of Ω . These are simply given by $G_{int}^{(DtI)} = T_{int}^{(DtN)} + ikI$ and $G_{ext}^{(DtI)} = T_{ext}^{(DtN)} + ikI$, respectively. It follows that

$$\partial_n \gamma_s|_{\partial\Omega} + ik\gamma_s|_{\partial\Omega} = G_{ext}^{(DtI)} \gamma_s|_{\partial\Omega} \quad (13)$$

$$\partial_n \beta_s|_{\partial\Omega} + ik\beta_s|_{\partial\Omega} = G_{int}^{(DtI)} \beta_s|_{\partial\Omega}. \quad (14)$$

So, we can rewrite the continuity conditions as

$$\partial_n \gamma_s|_{\partial\Omega} + ik\gamma_s|_{\partial\Omega} = G_{ext}^{(DtI)} \gamma_s|_{\partial\Omega} \quad (15)$$

$$= G_{ext}^{(DtI)} (\beta_s^{(h)}|_{\partial\Omega} + \beta_s^{(p)}|_{\partial\Omega}) \quad \text{by Equation (10)} \quad (16)$$

$$= G_{int}^{(DtI)} \beta_s^{(h)}|_{\partial\Omega} \quad \text{by Equation (12)} \quad (17)$$

which can be rearranged to set up a boundary integral equation (analogous to equation 1.14 of [Gillman et al. \(2015\)](#)):

$$(G_{int}^{(DtI)} - G_{ext}^{(DtI)}) \beta_s^{(h)}|_{\partial\Omega} = G_{ext}^{(DtI)} \beta_s^{(p)}|_{\partial\Omega}. \quad (18)$$

To put this all together, we can compute $\beta_s^{(p)}|_{\partial\Omega}$ as the impedance value from an upward HPS pass, and we can solve the boundary integral equation in Equation (18) to find $\beta_s^{(h)}|_{\partial\Omega}$. We can find $\beta_s|_{\Omega}$ throughout the scattering domain with a downward HPS pass, and from there we have $w_s|_{\Omega} = \alpha_s|_{\Omega} + \beta_s|_{\Omega}$ and can find $(DF[q]^* \xi)|_{\Omega}$ with Equation (4).

3 Jacobian-vector product

We briefly outline the procedure to compute $D\mathcal{F}_{k,s}[q]\delta q_s$ as well. See Theorem 3.1 of [Borges et al. \(2017\)](#) or the `derivative_solver.py` file for additional details. Similarly to the case of the vector-Jacobian case, we use the field $u_s^{(tot)}(x) = u_s^{(in)}(x) + u_s^{(sc)}(x)$ and set up an alternate PDE.

For each source, the Jacobian-vector product is

$$(D\mathcal{F}_{k,s}[q]\delta q_s)(r) = v_s(Rr), \quad (19)$$

where $R \in \mathbb{R}$ is the radius of the receiver ring and $r \in \mathbb{S}^1$ is the direction of a receiver. $v_s(x)$ is the solution to a PDE given by [Borges et al. \(2017, Theorem 3.1\)](#):

$$\begin{cases} \Delta v_s(x) + k^2(q(x) + 1)v_s(x) = -k^2\delta q(x)u_s^{(tot)}(x), & x \in \mathbb{R}^2; \\ \frac{\partial}{\partial r} v_s(x) - ikv_s(x) = o(r^{-1/2}), & \text{as } r = \|x\|_2 \rightarrow \infty. \end{cases} \quad (20)$$

This is very similar to Equation (6), except with $\delta q(x)u_s^{(tot)}(x)$ in place of $q(x)\alpha_s(x)$ and the standard Sommerfeld radiation condition rather than the adjoint one. As such, $v_s(x)$ can be solved in a manner similar to $\beta_s(x)$ in Section 2.

To apply the Jacobian for all sources at once, each source can be treated independently:

$$D\mathcal{F}_k[q]\xi = D\mathcal{F}_k[q] \begin{bmatrix} \xi_{s_1} \\ \xi_{s_2} \\ \vdots \\ \xi_{s_{N_s}} \end{bmatrix} = \begin{bmatrix} D\mathcal{F}_{k,s_1}[q]\xi_{s_1} \\ D\mathcal{F}_{k,s_2}[q]\xi_{s_2} \\ \vdots \\ D\mathcal{F}_{k,s_{N_s}}[q]\xi_{s_{N_s}} \end{bmatrix}. \quad (21)$$

References

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